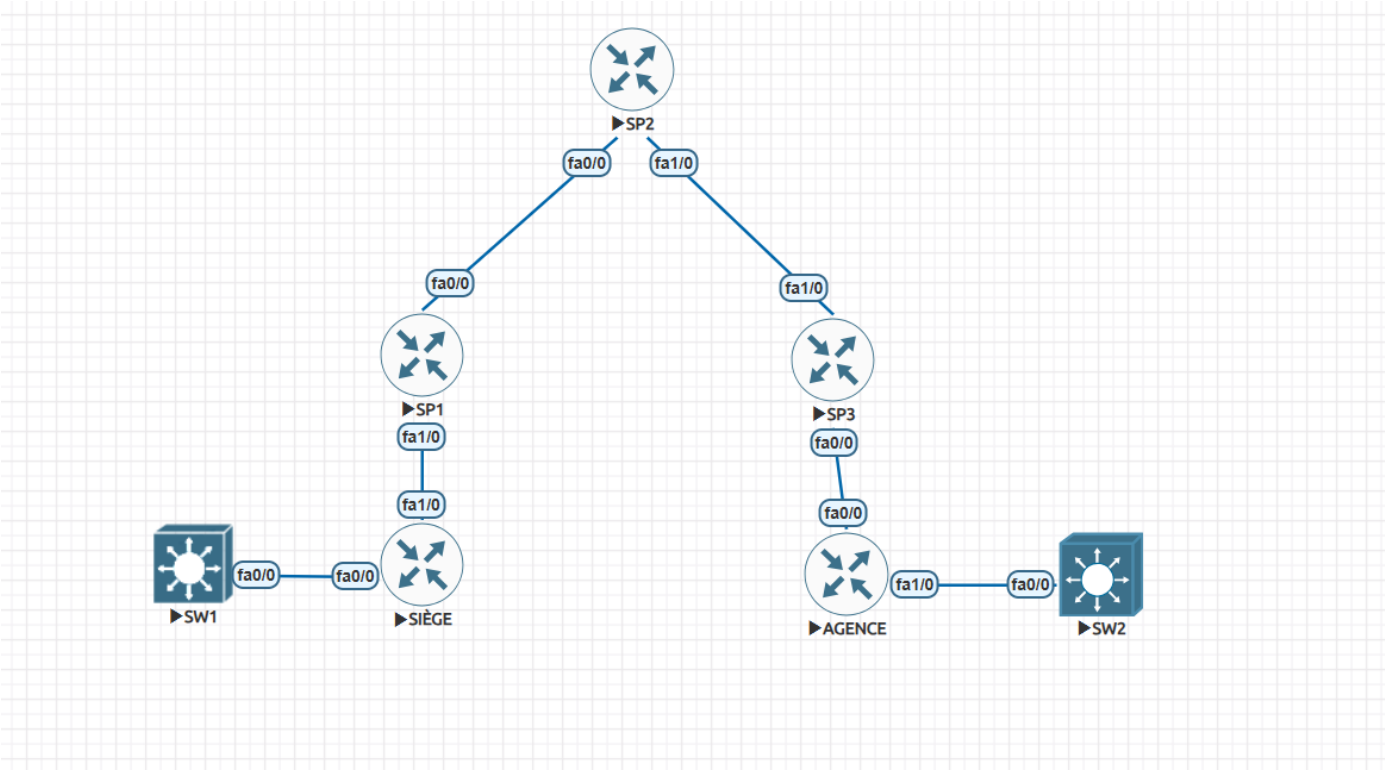
TP-MPLS-1 : Configuration d'un backbone MPLS (partie 1)

Malek Boughdiri RSI31

Ma Topologie :

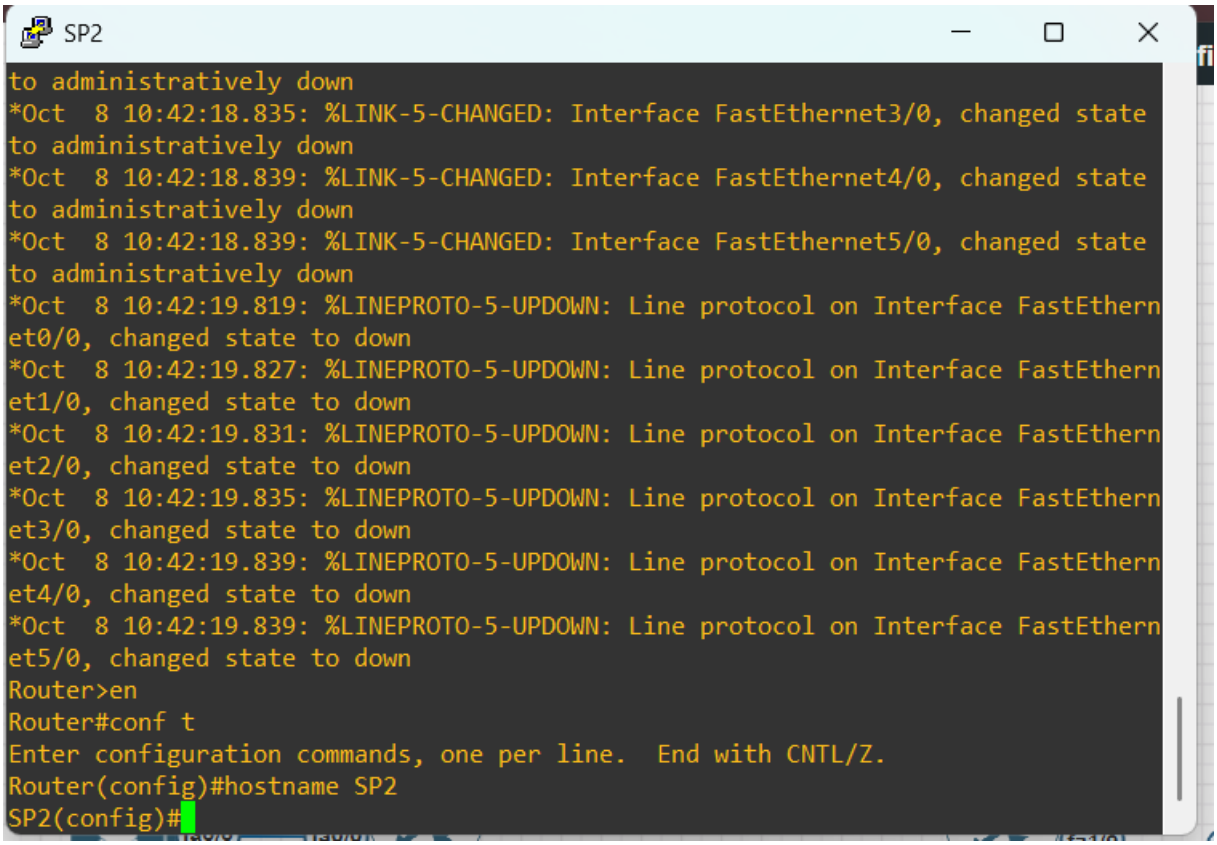


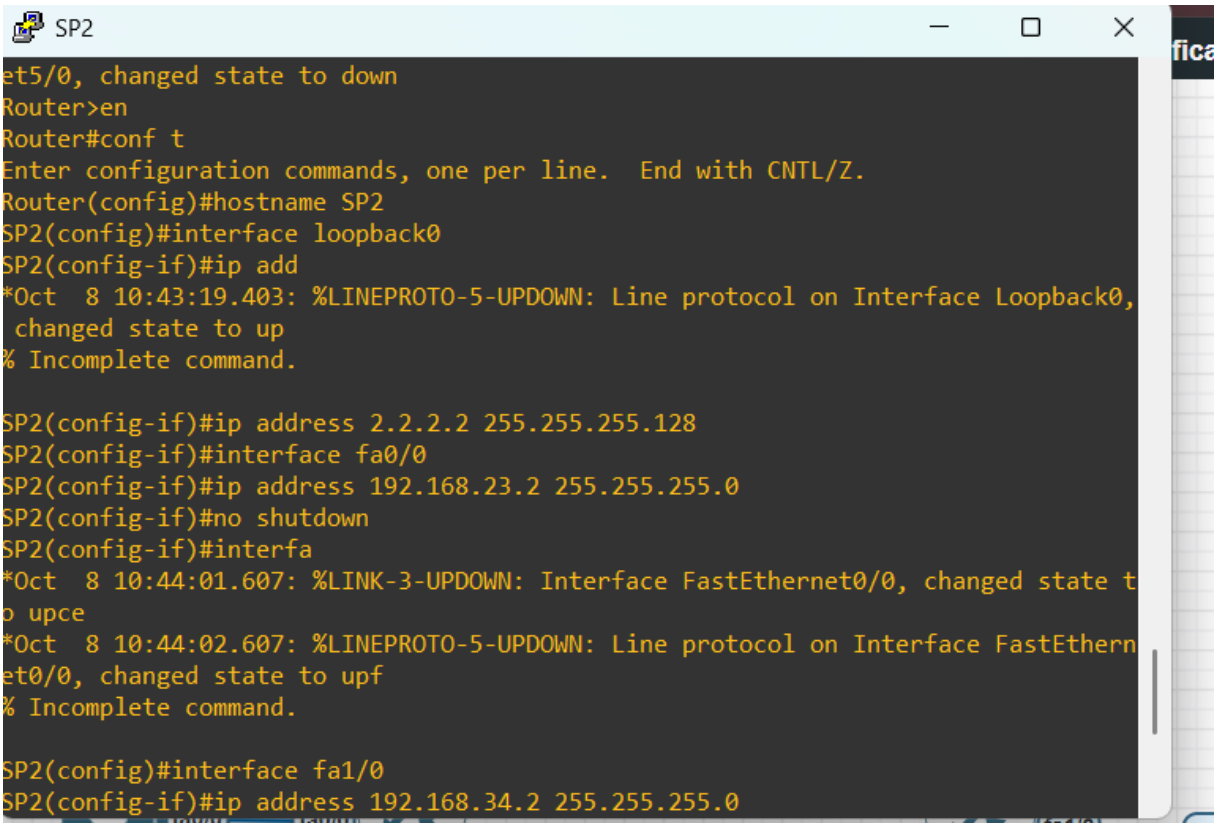
Étape 1 : Configuration de base

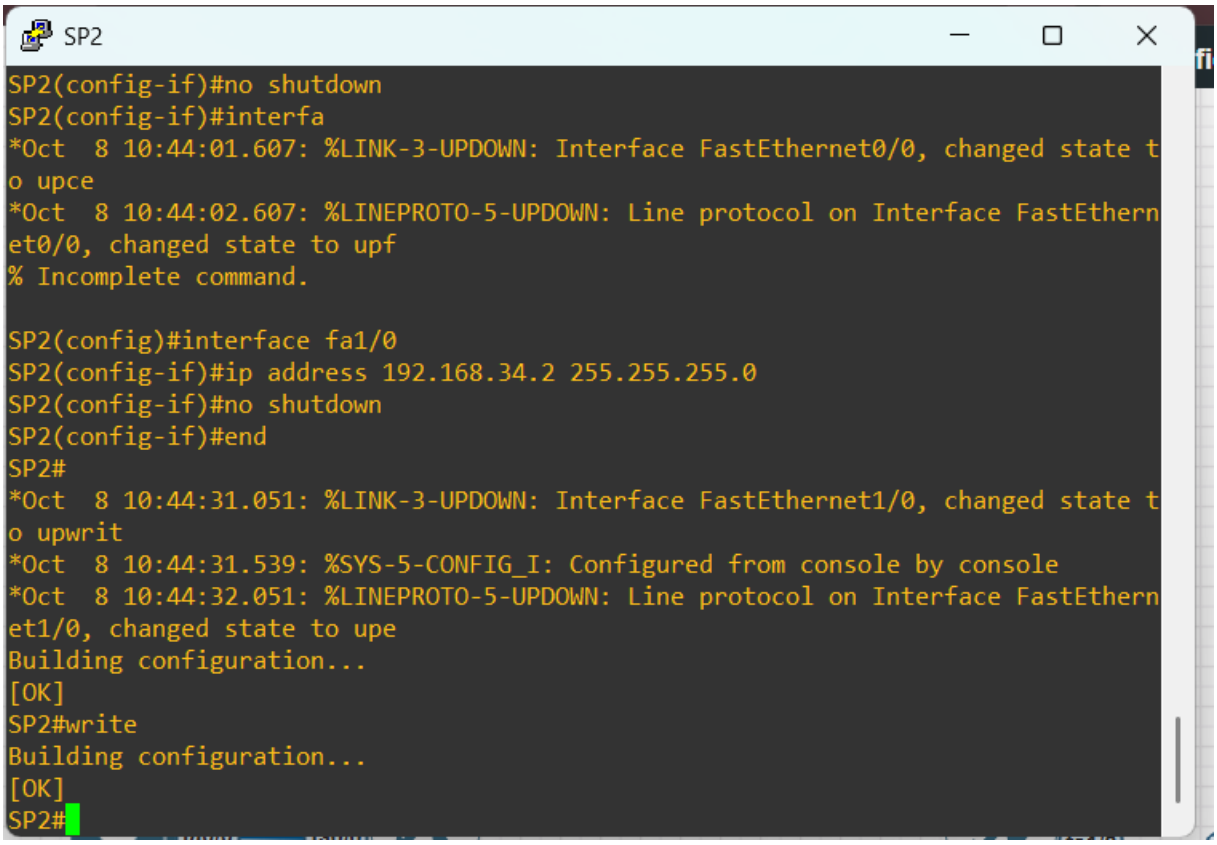
SP1

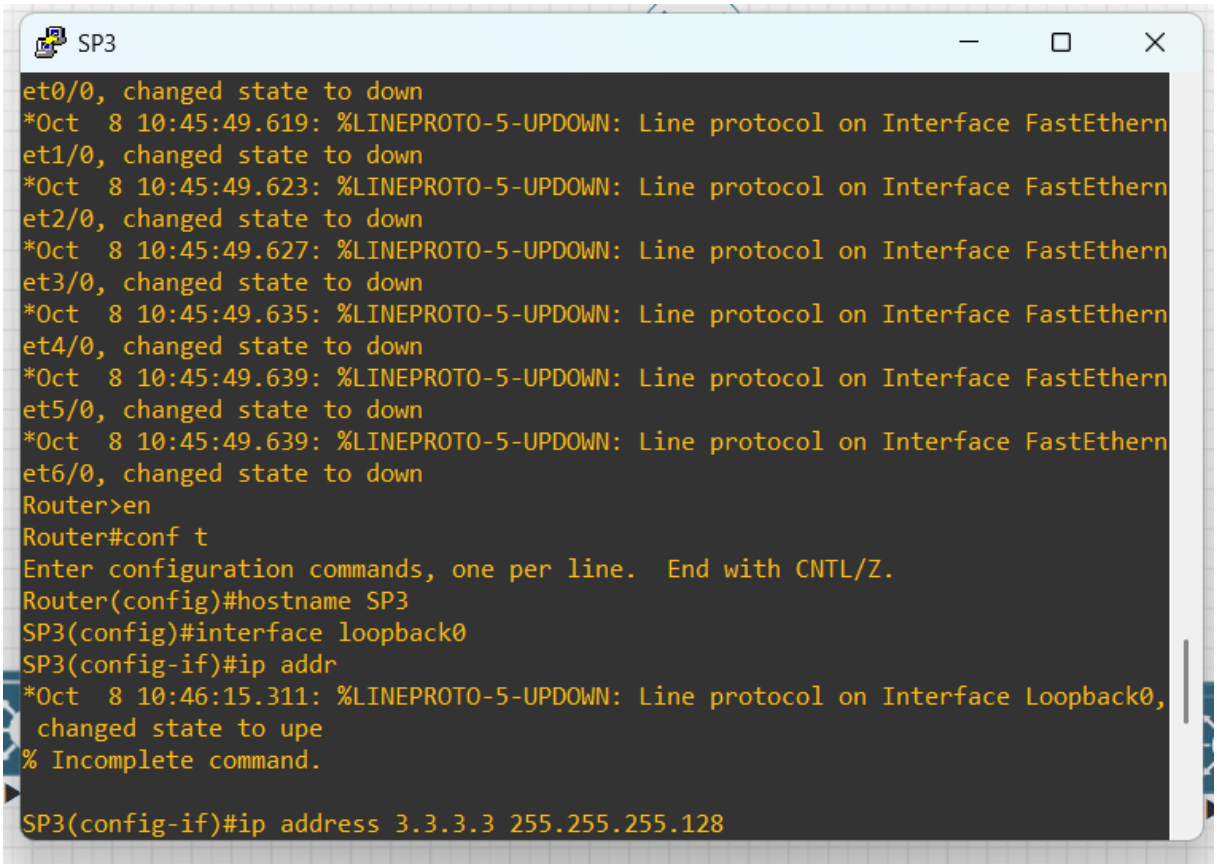
```
SP1(config-if)#interface fa0/0
SP1(config-if)#ip address 192.168.23.1 255.255.255.0
SP1(config-if)#no shutdown
SP1(config-if)#interface fa
*Oct  8 10:40:14.935: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Oct  8 10:40:15.935: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
% Incomplete command.

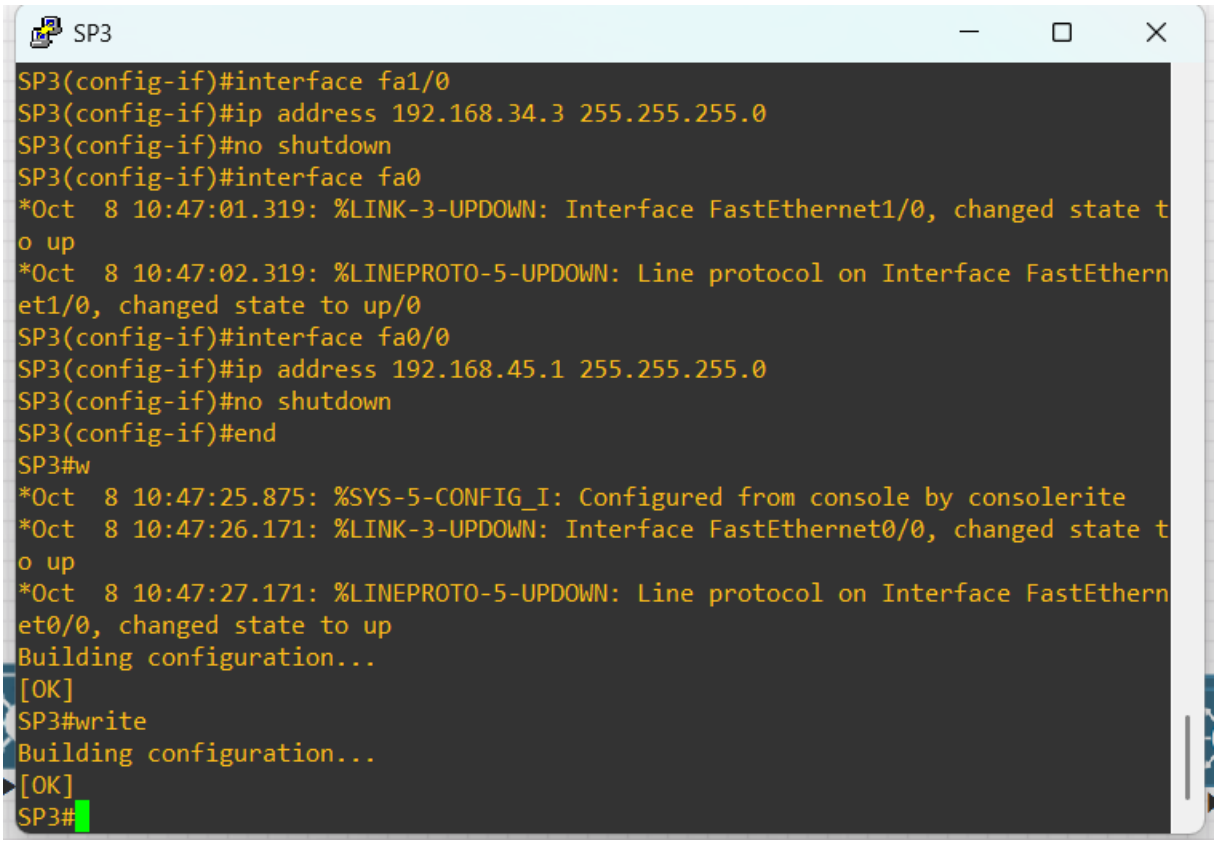
SP1(config)#interface fa1/0
SP1(config-if)#ip address 192.168.12.1 255.255.255.0
SP1(config-if)#no shutdown
SP1(config-if)#end
SP1#
*Oct  8 10:40:46.795: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state t
o up
*Oct  8 10:40:46.943: %SYS-5-CONFIG_I: Configured from console by console
*Oct  8 10:40:47.795: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et1/0, changed state to up
SP1#write
Building configuration...
[OK]
SP1#
```

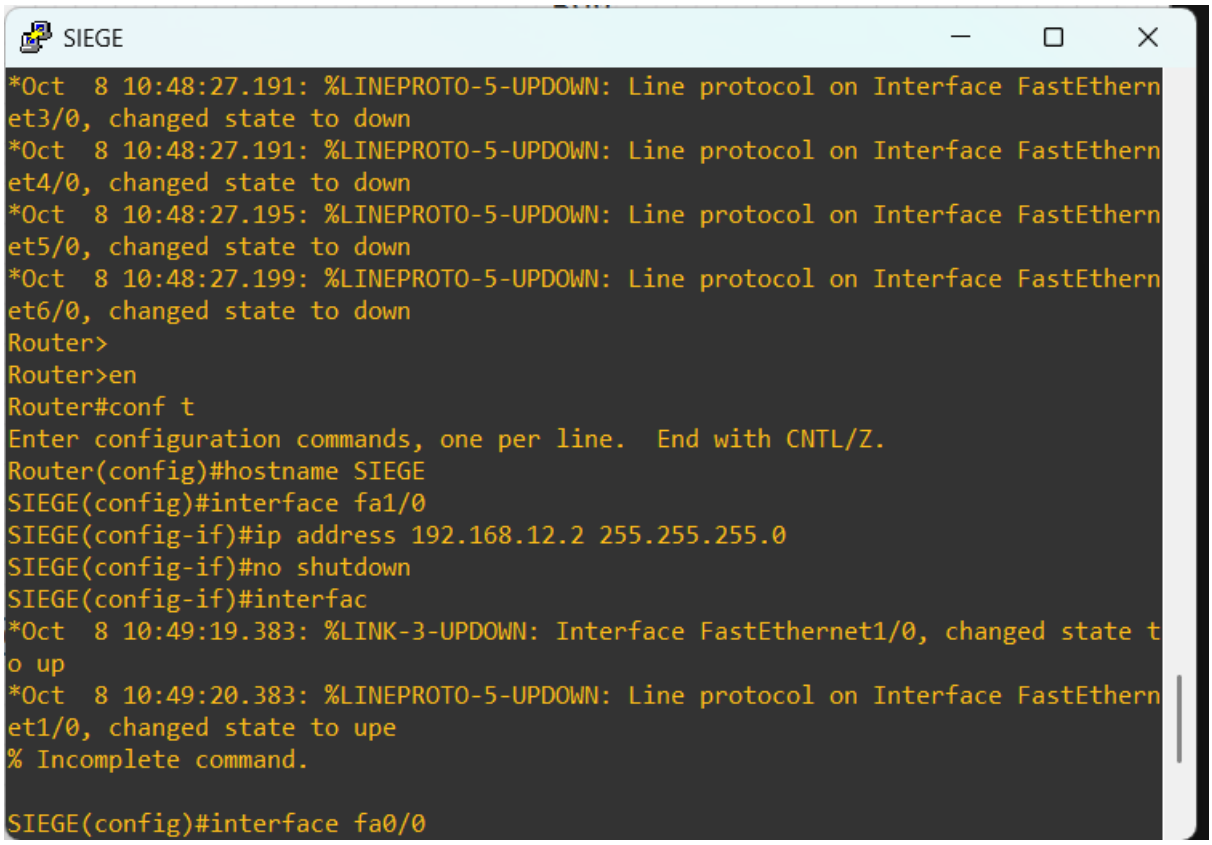

SP2
to administratively down
*Oct 8 10:42:18.835: %LINK-5-CHANGED: Interface FastEthernet3/0, changed state to administratively down
*Oct 8 10:42:18.839: %LINK-5-CHANGED: Interface FastEthernet4/0, changed state to administratively down
*Oct 8 10:42:18.839: %LINK-5-CHANGED: Interface FastEthernet5/0, changed state to administratively down
*Oct 8 10:42:19.819: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to down
*Oct 8 10:42:19.827: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to down
*Oct 8 10:42:19.831: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to down
*Oct 8 10:42:19.835: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet3/0, changed state to down
*Oct 8 10:42:19.839: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet4/0, changed state to down
*Oct 8 10:42:19.839: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet5/0, changed state to down
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname SP2
SP2(config)#

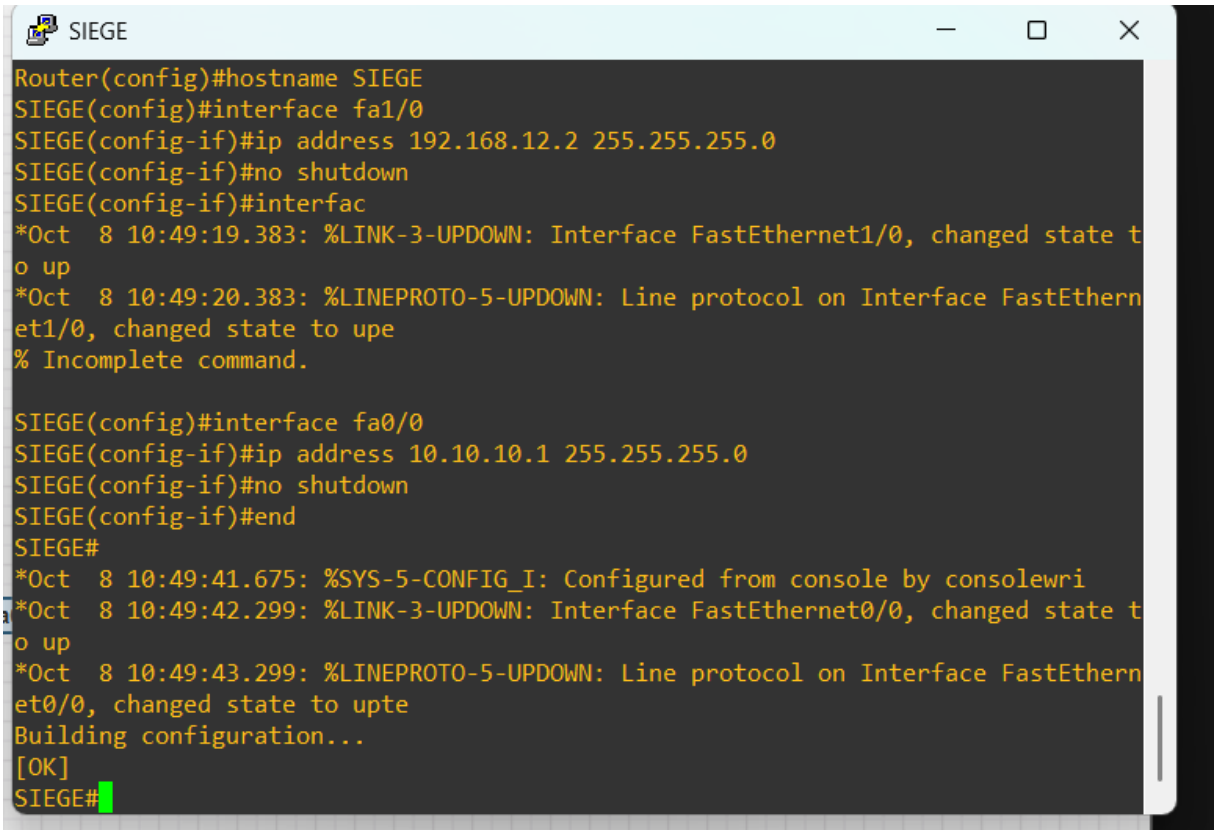

SP2
et5/0, changed state to down
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname SP2
SP2(config)#interface loopback0
SP2(config-if)#ip add
*Oct 8 10:43:19.403: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
% Incomplete command.
SP2(config-if)#ip address 2.2.2.2 255.255.255.128
SP2(config-if)#interface fa0/0
SP2(config-if)#ip address 192.168.23.2 255.255.255.0
SP2(config-if)#no shutdown
SP2(config-if)#interfa
*Oct 8 10:44:01.607: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to upce
*Oct 8 10:44:02.607: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to upf
% Incomplete command.
SP2(config)#interface fa1/0
SP2(config-if)#ip address 192.168.34.2 255.255.255.0

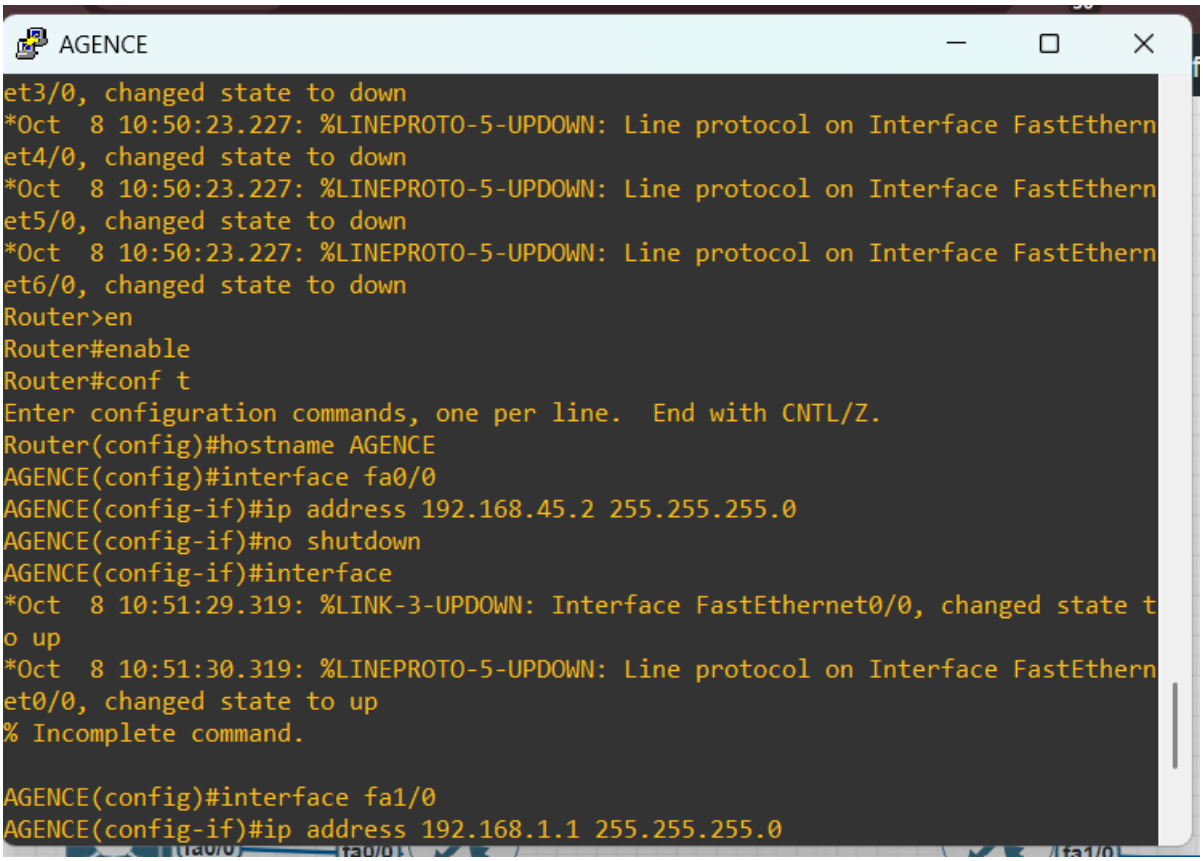

SP2
SP2(config-if)#no shutdown
SP2(config-if)#interfa
*Oct 8 10:44:01.607: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o upce
*Oct 8 10:44:02.607: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to upf
% Incomplete command.
SP2(config)#interface fa1/0
SP2(config-if)#ip address 192.168.34.2 255.255.255.0
SP2(config-if)#no shutdown
SP2(config-if)#end
SP2#
*Oct 8 10:44:31.051: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state t
o upwrit
*Oct 8 10:44:31.539: %SYS-5-CONFIG_I: Configured from console by console
*Oct 8 10:44:32.051: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et1/0, changed state to upe
Building configuration...
[OK]
SP2#write
Building configuration...
[OK]
SP2#

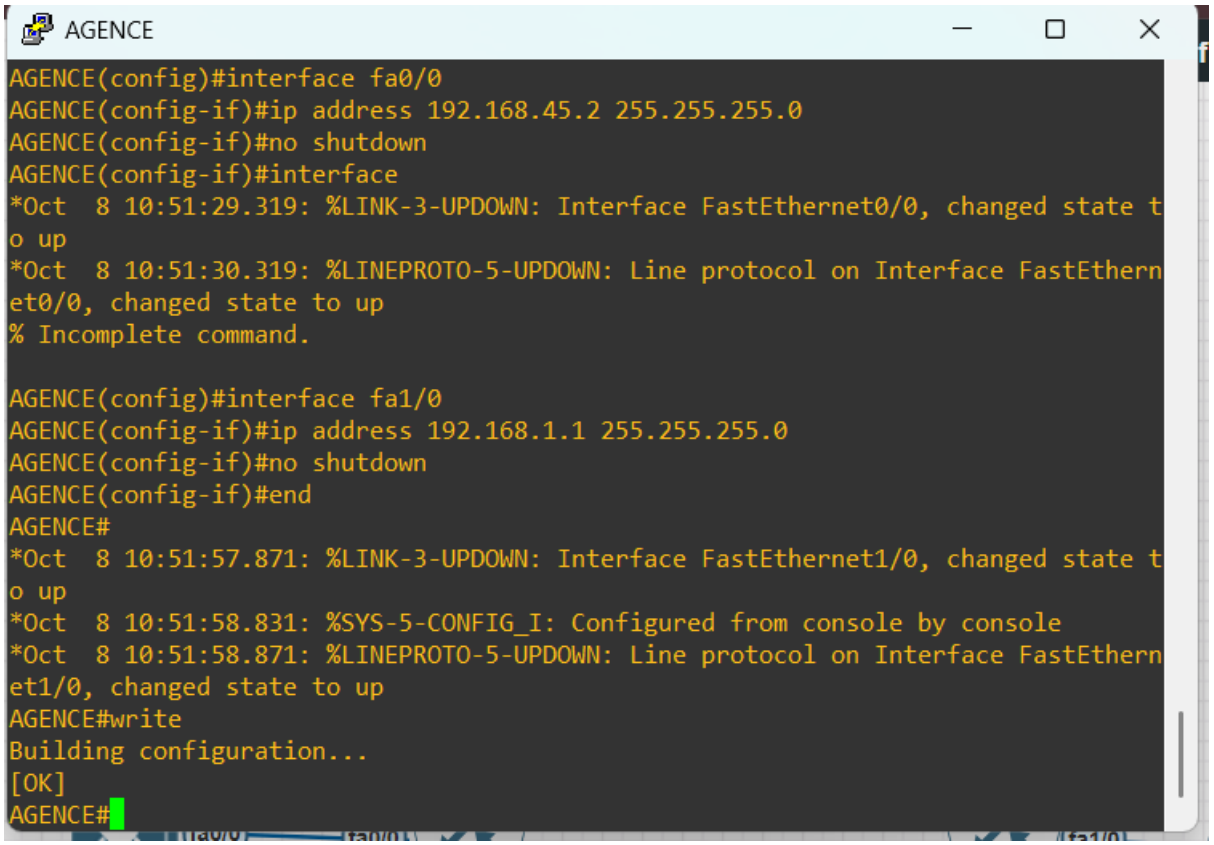

SP3
et0/0, changed state to down
*Oct 8 10:45:49.619: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et1/0, changed state to down
*Oct 8 10:45:49.623: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et2/0, changed state to down
*Oct 8 10:45:49.627: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et3/0, changed state to down
*Oct 8 10:45:49.635: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et4/0, changed state to down
*Oct 8 10:45:49.639: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et5/0, changed state to down
*Oct 8 10:45:49.639: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et6/0, changed state to down
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname SP3
SP3(config)#interface loopback0
SP3(config-if)#ip addr
*Oct 8 10:46:15.311: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0,
 changed state to upe
% Incomplete command.
SP3(config-if)#ip address 3.3.3.3 255.255.255.128

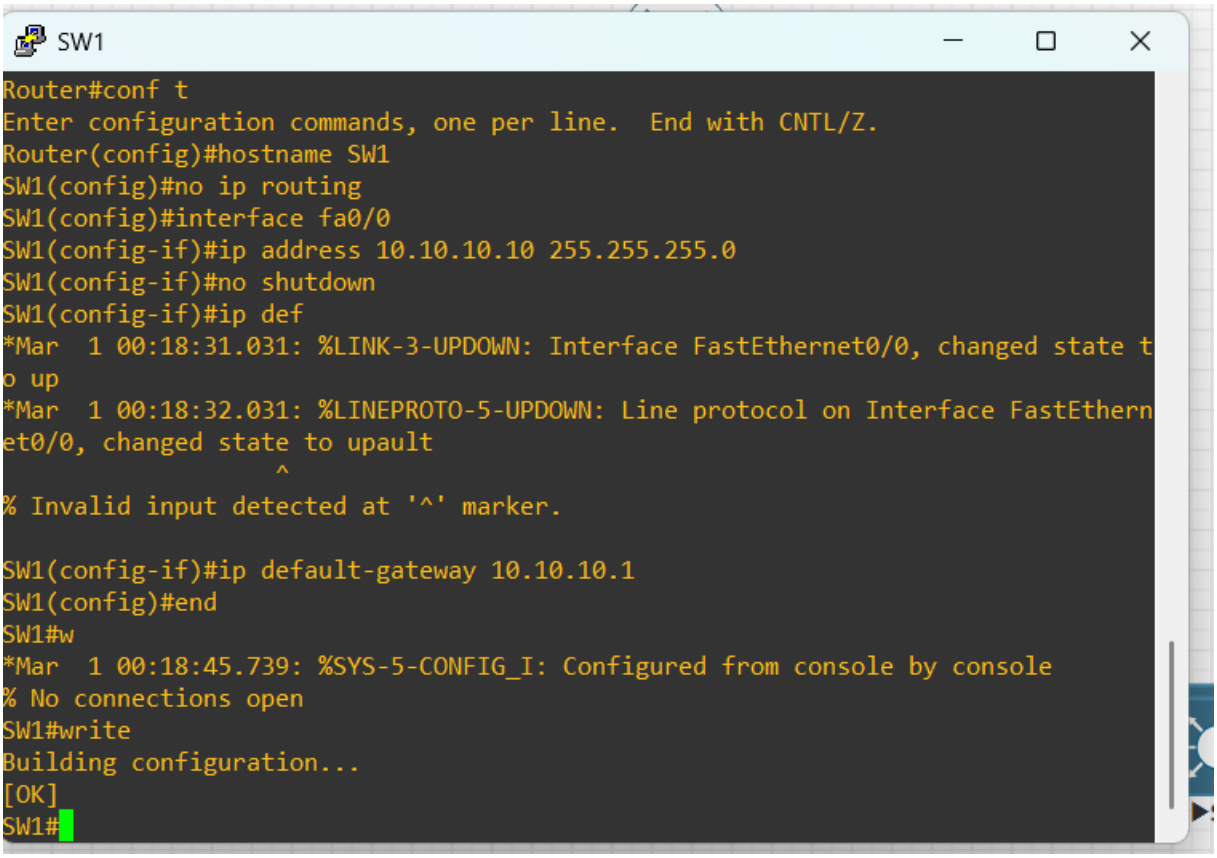

SP3
SP3(config-if)#interface fa1/0
SP3(config-if)#ip address 192.168.34.3 255.255.255.0
SP3(config-if)#no shutdown
SP3(config-if)#interface fa0
*Oct 8 10:47:01.319: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state t
o up
*Oct 8 10:47:02.319: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et1/0, changed state to up/0
SP3(config-if)#interface fa0/0
SP3(config-if)#ip address 192.168.45.1 255.255.255.0
SP3(config-if)#no shutdown
SP3(config-if)#end
SP3#w
*Oct 8 10:47:25.875: %SYS-5-CONFIG_I: Configured from console by consolerite
*Oct 8 10:47:26.171: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Oct 8 10:47:27.171: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
Building configuration...
[OK]
SP3#write
Building configuration...
[OK]
SP3#

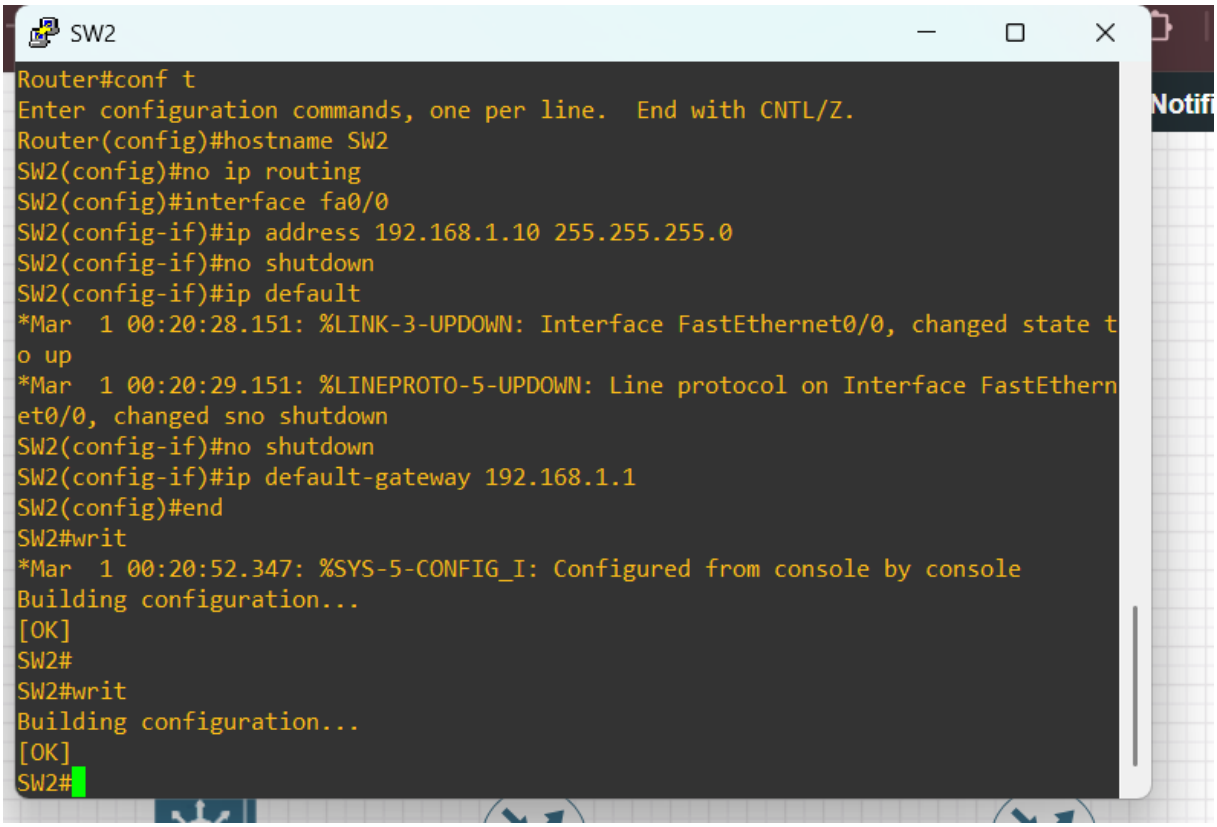

SIEGE
*Oct 8 10:48:27.191: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et3/0, changed state to down
*Oct 8 10:48:27.191: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et4/0, changed state to down
*Oct 8 10:48:27.195: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et5/0, changed state to down
*Oct 8 10:48:27.199: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et6/0, changed state to down
Router>
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname SIEGE
SIEGE(config)#interface fa1/0
SIEGE(config-if)#ip address 192.168.12.2 255.255.255.0
SIEGE(config-if)#no shutdown
SIEGE(config-if)#interfac
*Oct 8 10:49:19.383: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state t
o up
*Oct 8 10:49:20.383: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et1/0, changed state to upe
% Incomplete command.
SIEGE(config)#interface fa0/0


SIEGE
Router(config)#hostname SIEGE
SIEGE(config)#interface fa1/0
SIEGE(config-if)#ip address 192.168.12.2 255.255.255.0
SIEGE(config-if)#no shutdown
SIEGE(config-if)#interfac
*Oct 8 10:49:19.383: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state t
o up
*Oct 8 10:49:20.383: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et1/0, changed state to upe
% Incomplete command.
SIEGE(config)#interface fa0/0
SIEGE(config-if)#ip address 10.10.10.1 255.255.255.0
SIEGE(config-if)#no shutdown
SIEGE(config-if)#end
SIEGE#
*Oct 8 10:49:41.675: %SYS-5-CONFIG_I: Configured from console by consolewri
*Oct 8 10:49:42.299: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Oct 8 10:49:43.299: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to upte
Building configuration...
[OK]
SIEGE#

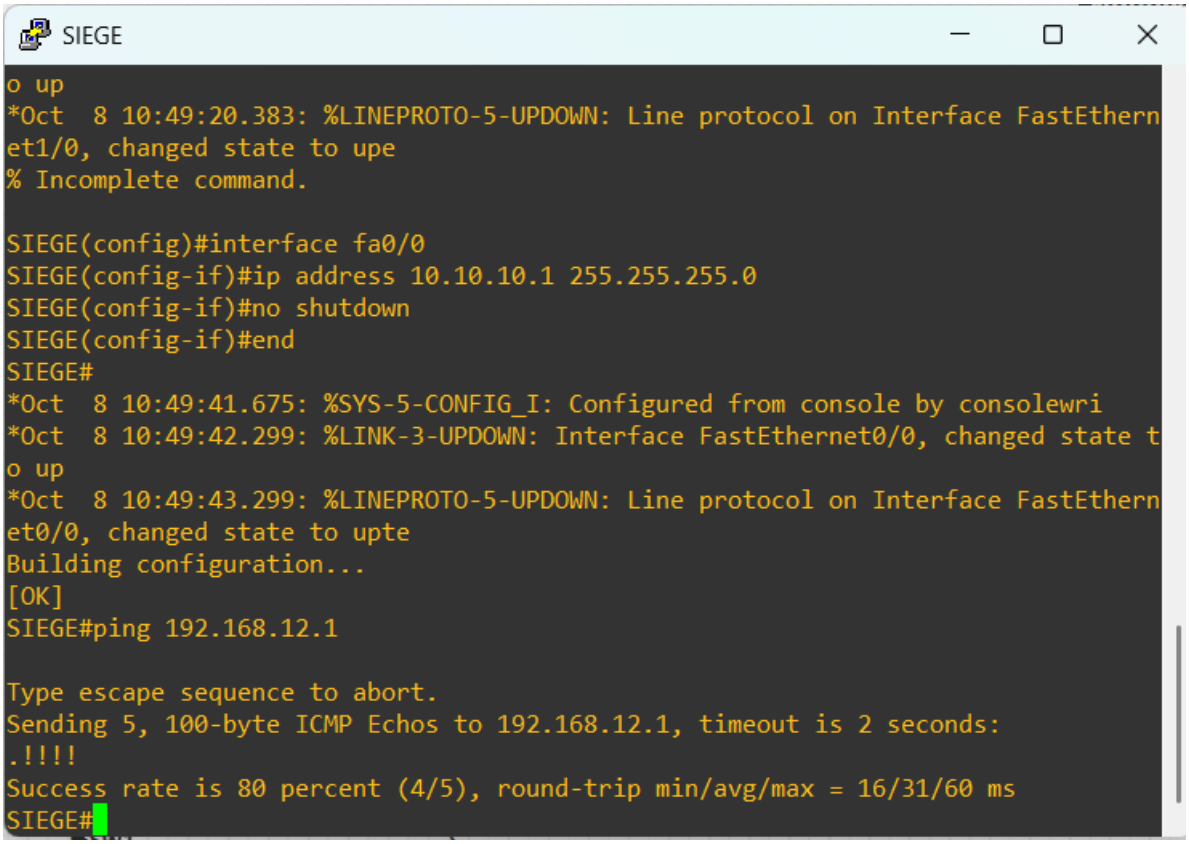

AGENCE
et3/0, changed state to down
*Oct 8 10:50:23.227: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et4/0, changed state to down
*Oct 8 10:50:23.227: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et5/0, changed state to down
*Oct 8 10:50:23.227: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et6/0, changed state to down
Router>en
Router#enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname AGENCE
AGENCE(config)#interface fa0/0
AGENCE(config-if)#ip address 192.168.45.2 255.255.255.0
AGENCE(config-if)#no shutdown
AGENCE(config-if)#interface
*Oct 8 10:51:29.319: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Oct 8 10:51:30.319: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
% Incomplete command.
AGENCE(config)#interface fa1/0
AGENCE(config-if)#ip address 192.168.1.1 255.255.255.0

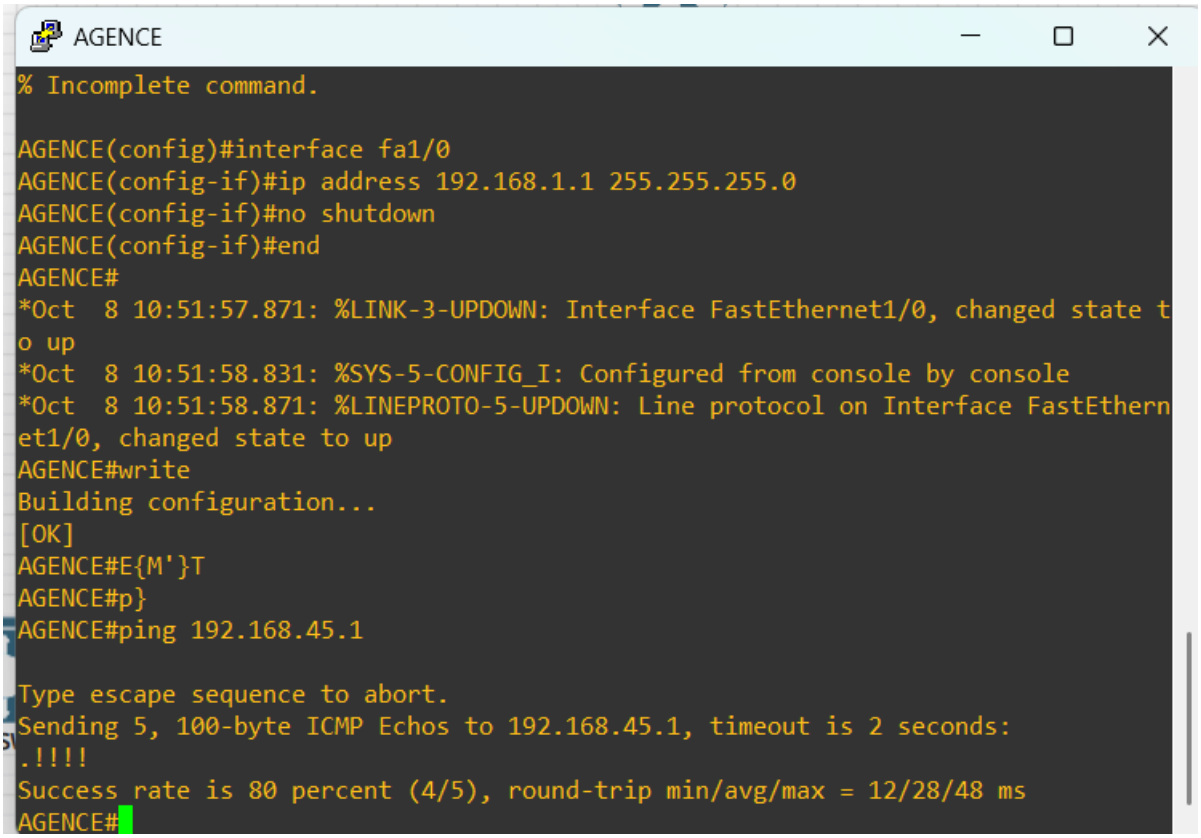

AGENCE
AGENCE(config)#interface fa0/0
AGENCE(config-if)#ip address 192.168.45.2 255.255.255.0
AGENCE(config-if)#no shutdown
AGENCE(config-if)#interface
*Oct 8 10:51:29.319: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Oct 8 10:51:30.319: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
% Incomplete command.
AGENCE(config)#interface fa1/0
AGENCE(config-if)#ip address 192.168.1.1 255.255.255.0
AGENCE(config-if)#no shutdown
AGENCE(config-if)#end
AGENCE#
*Oct 8 10:51:57.871: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state t
o up
*Oct 8 10:51:58.831: %SYS-5-CONFIG_I: Configured from console by console
*Oct 8 10:51:58.871: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et1/0, changed state to up
AGENCE#write
Building configuration...
[OK]
AGENCE#

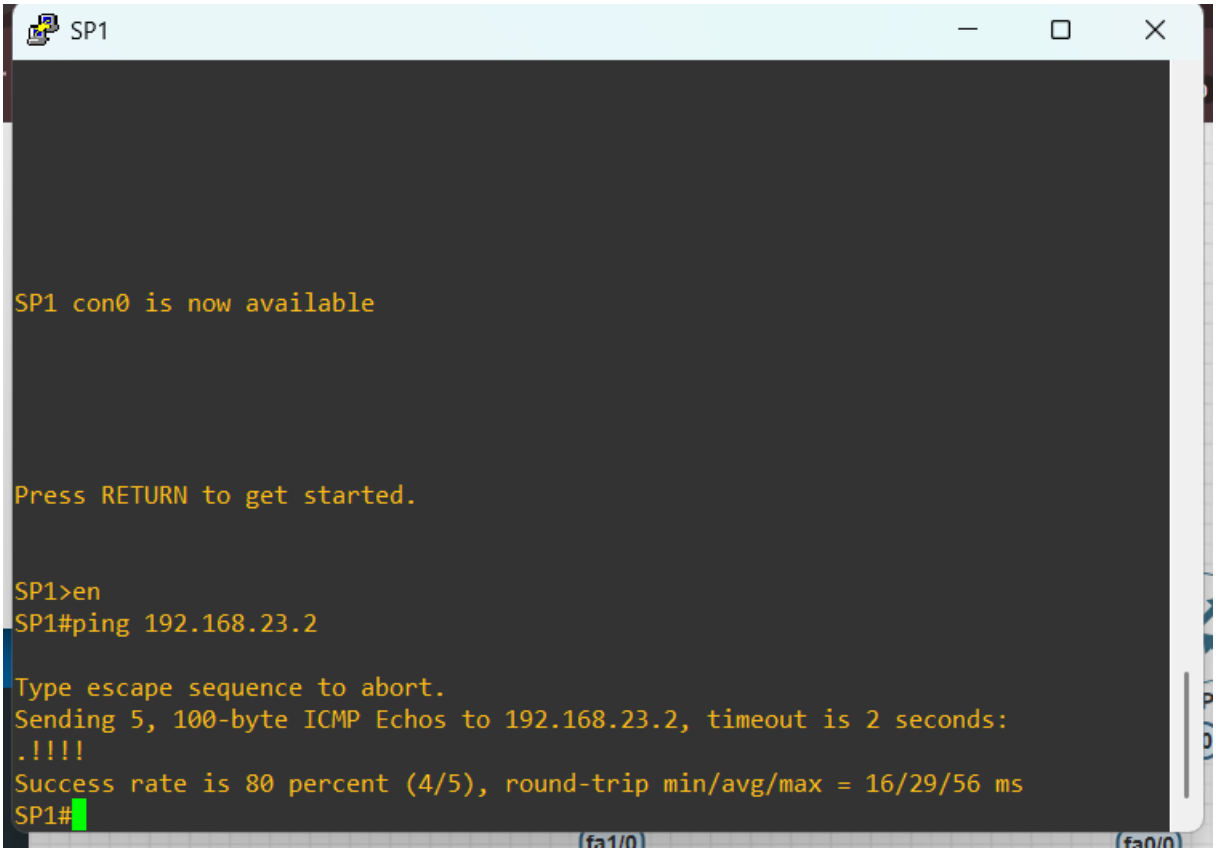

SW1
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname SW1
SW1(config)#no ip routing
SW1(config)#interface fa0/0
SW1(config-if)#ip address 10.10.10.10 255.255.255.0
SW1(config-if)#no shutdown
SW1(config-if)#ip def
*Mar 1 00:18:31.031: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar 1 00:18:32.031: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to upault
^
% Invalid input detected at '^' marker.
SW1(config-if)#ip default-gateway 10.10.10.1
SW1(config)#end
SW1#w
*Mar 1 00:18:45.739: %SYS-5-CONFIG_I: Configured from console by console
% No connections open
SW1#write
Building configuration...
[OK]
SW1#

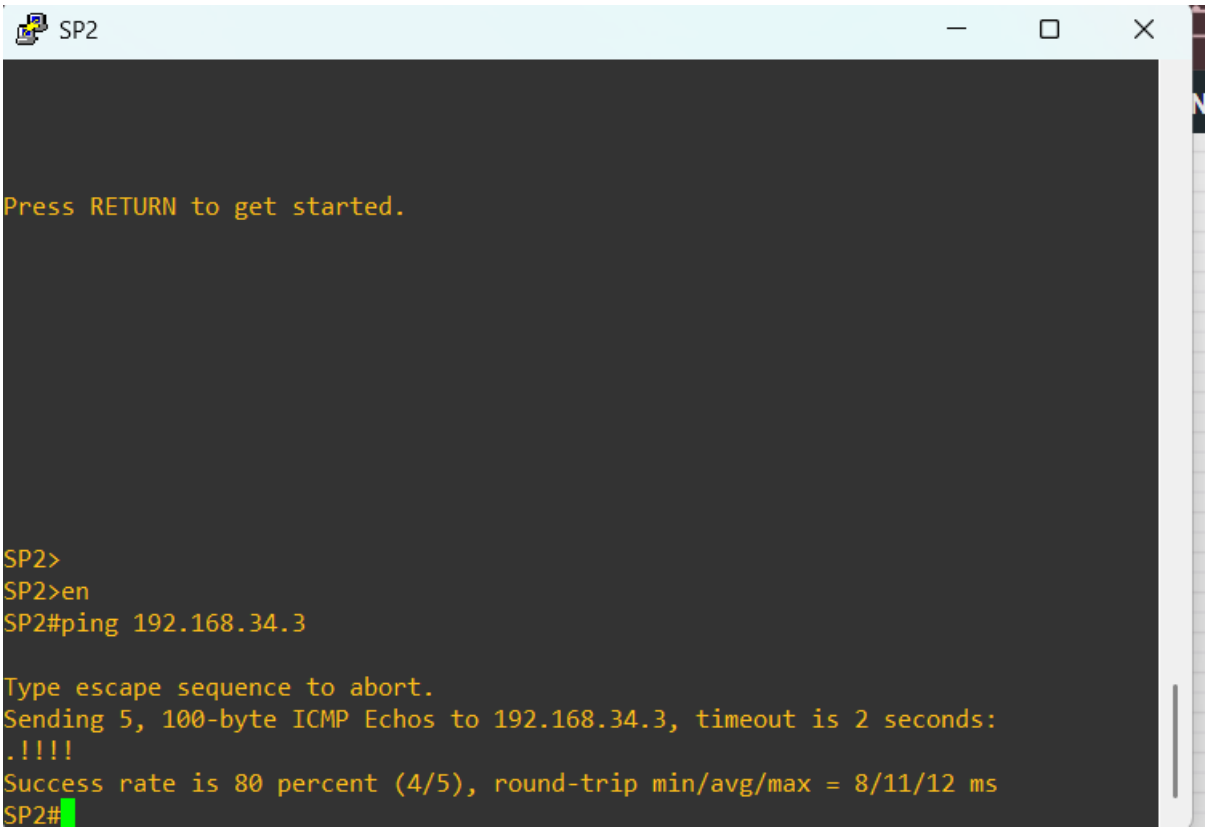

SW2
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname SW2
SW2(config)#no ip routing
SW2(config)#interface fa0/0
SW2(config-if)#ip address 192.168.1.10 255.255.255.0
SW2(config-if)#no shutdown
SW2(config-if)#ip default
*Mar 1 00:20:28.151: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar 1 00:20:29.151: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed sno shutdown
SW2(config-if)#no shutdown
SW2(config-if)#ip default-gateway 192.168.1.1
SW2(config)#end
SW2#writ
*Mar 1 00:20:52.347: %SYS-5-CONFIG_I: Configured from console by console
Building configuration...
[OK]
SW2#
SW2#writ
Building configuration...
[OK]
SW2#

PING :

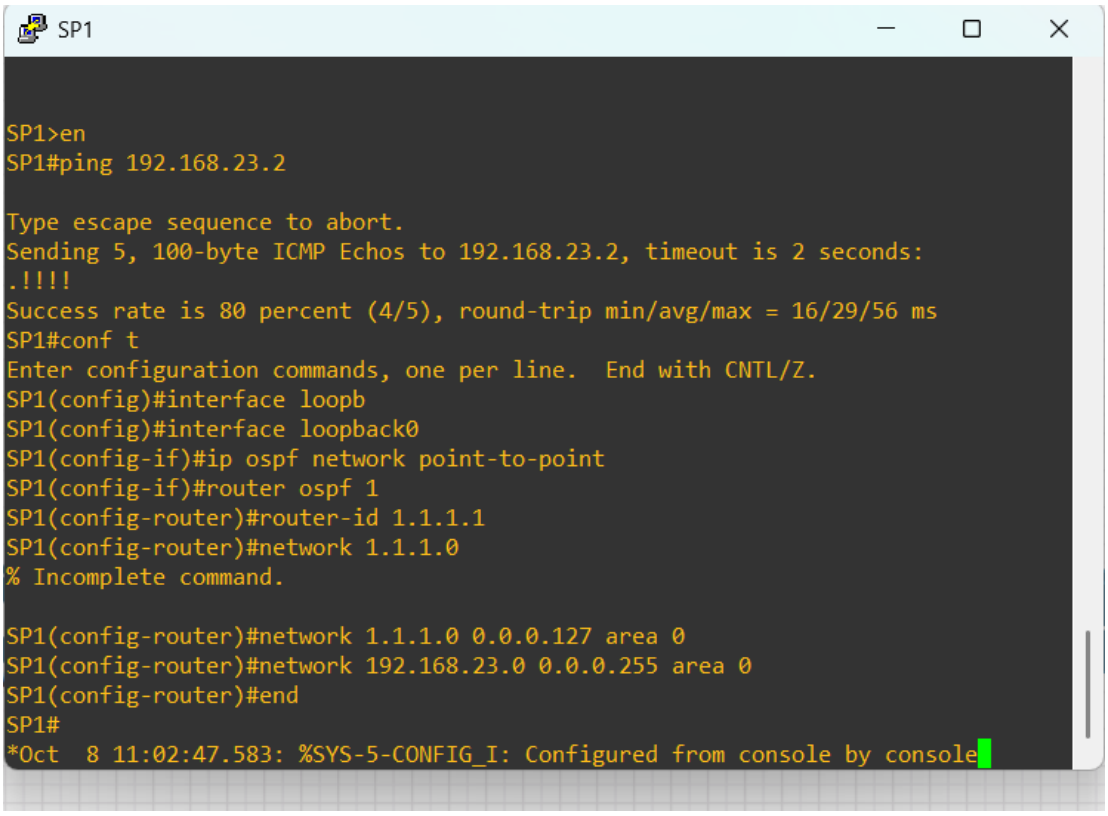

SIEGE
o up
*Oct 8 10:49:20.383: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et1/0, changed state to upe
% Incomplete command.
SIEGE(config)#interface fa0/0
SIEGE(config-if)#ip address 10.10.10.1 255.255.255.0
SIEGE(config-if)#no shutdown
SIEGE(config-if)#end
SIEGE#
*Oct 8 10:49:41.675: %SYS-5-CONFIG_I: Configured from console by consolewri
*Oct 8 10:49:42.299: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Oct 8 10:49:43.299: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to upte
Building configuration...
[OK]
SIEGE#ping 192.168.12.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 16/31/60 ms
SIEGE#

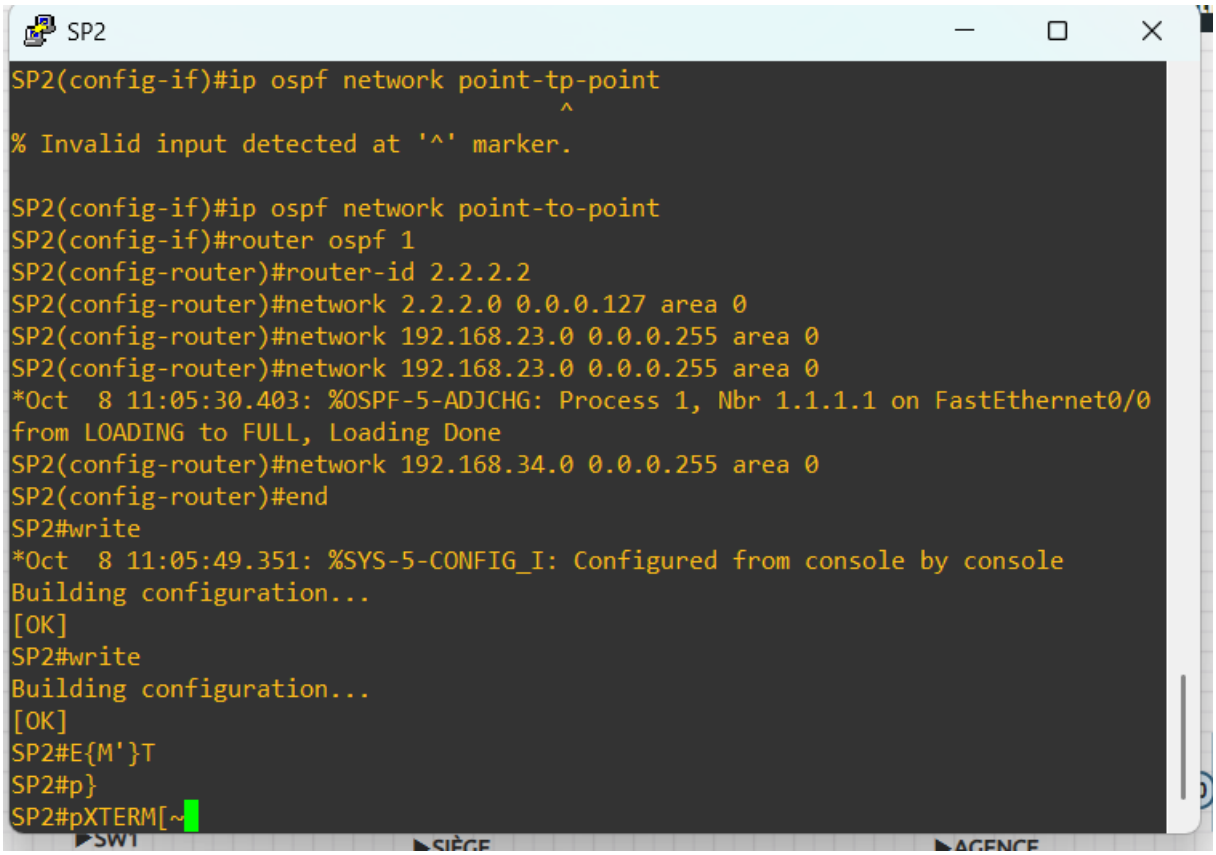

AGENCE
% Incomplete command.
AGENCE(config)#interface fa1/0
AGENCE(config-if)#ip address 192.168.1.1 255.255.255.0
AGENCE(config-if)#no shutdown
AGENCE(config-if)#end
AGENCE#
*Oct 8 10:51:57.871: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state t
o up
*Oct 8 10:51:58.831: %SYS-5-CONFIG_I: Configured from console by console
*Oct 8 10:51:58.871: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et1/0, changed state to up
AGENCE#write
Building configuration...
[OK]
AGENCE#E{M'}T
AGENCE#p}
AGENCE#ping 192.168.45.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.45.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 12/28/48 ms
AGENCE#

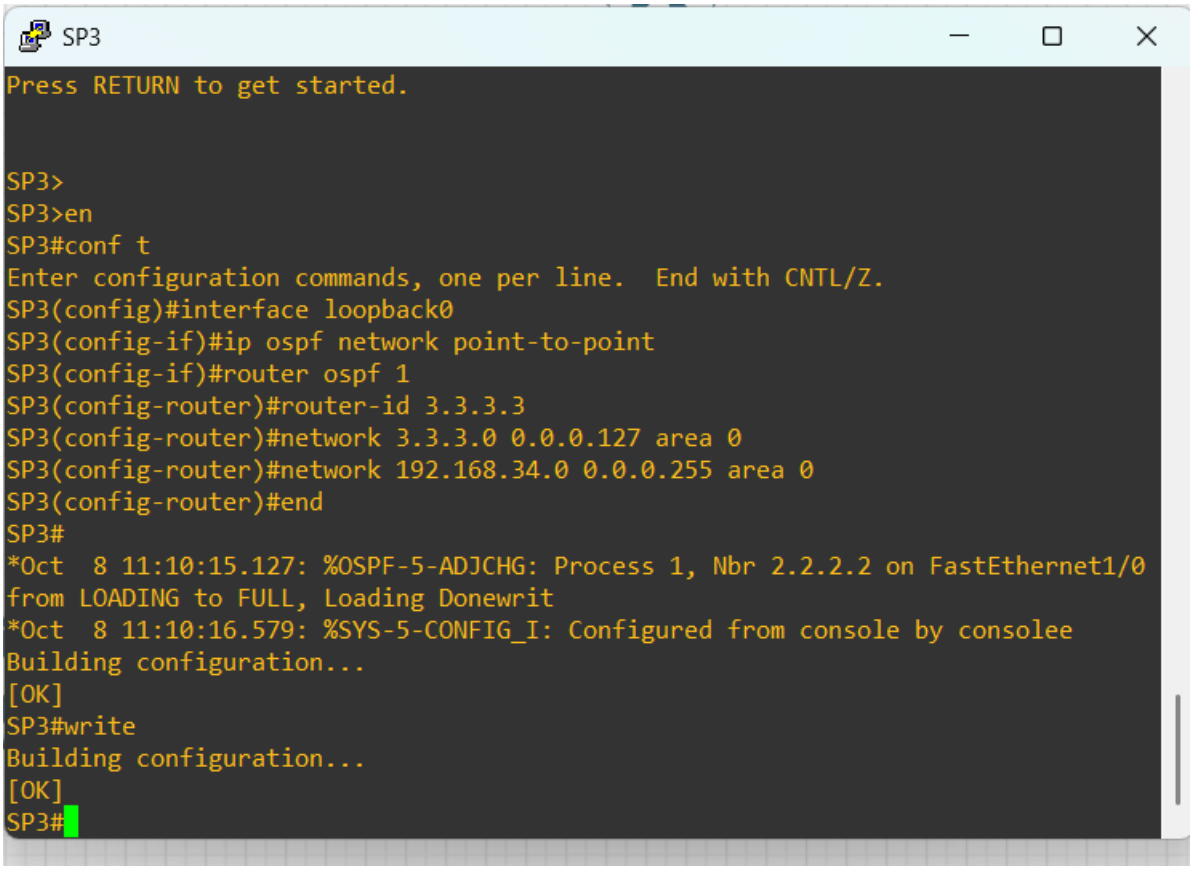

SP1
SP1 con0 is now available
Press RETURN to get started.
SP1>en
SP1#ping 192.168.23.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.23.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 16/29/56 ms
SP1#



Étape 2 : OSPF dans le backbone (Area 0)




SP2
SP2(config-if)#ip ospf network point-tp-point
^
% Invalid input detected at '^' marker.
SP2(config-if)#ip ospf network point-to-point
SP2(config-if)#router ospf 1
SP2(config-router)#router-id 2.2.2.2
SP2(config-router)#network 2.2.2.0 0.0.0.127 area 0
SP2(config-router)#network 192.168.23.0 0.0.0.255 area 0
SP2(config-router)#network 192.168.23.0 0.0.0.255 area 0
*Oct 8 11:05:30.403: %OSPF-5-ADJCHG: Process 1, Nbr 1.1.1.1 on FastEthernet0/0
from LOADING to FULL, Loading Done
SP2(config-router)#network 192.168.34.0 0.0.0.255 area 0
SP2(config-router)#end
SP2#write
*Oct 8 11:05:49.351: %SYS-5-CONFIG_I: Configured from console by console
Building configuration...
[OK]
SP2#write
Building configuration...
[OK]
SP2#E{M'}T
SP2#p}
SP2#pXTERM[~


SP3
Press RETURN to get started.
SP3>
SP3>en
SP3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SP3(config)#interface loopback0
SP3(config-if)#ip ospf network point-to-point
SP3(config-if)#router ospf 1
SP3(config-router)#router-id 3.3.3.3
SP3(config-router)#network 3.3.3.0 0.0.0.127 area 0
SP3(config-router)#network 192.168.34.0 0.0.0.255 area 0
SP3(config-router)#end
SP3#
*Oct 8 11:10:15.127: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on FastEthernet1/0
from LOADING to FULL, Loading Donewrit
*Oct 8 11:10:16.579: %SYS-5-CONFIG_I: Configured from console by consolee
Building configuration...
[OK]
SP3#write
Building configuration...
[OK]
SP3#

Vérification :

SP1

```
SP1#
SP1#
SP1#
SP1#
SP1#write
Building configuration...
[OK]
SP1#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

      2.0.0.0/25 is subnetted, 1 subnets
O        2.2.2.0 [110/2] via 192.168.23.2, 00:05:35, FastEthernet0/0
      3.0.0.0/25 is subnetted, 1 subnets
O        3.3.3.0 [110/3] via 192.168.23.2, 00:00:40, FastEthernet0/0
O     192.168.34.0/24 [110/2] via 192.168.23.2, 00:05:19, FastEthernet0/0
SP1#
```

SP2

```
Building configuration...
[OK]
SP2#E{M'}T
SP2#p}
SP2#
*Oct  8 11:10:14.807: %OSPF-5-ADJCHG: Process 1, Nbr 3.3.3.3 on FastEthernet1/0
from LOADING to FULL, Loading DoneMMpppTERM
Translating "MMpppTERM"...domain server (255.255.255.255)

% Bad IP address or host name
Translating "MMpppTERM"...domain server (255.255.255.255)
 (255.255.255.255)
Translating "MMpppTERM"...domain server (255.255.255.255)

% Unknown command or computer name, or unable to find computer address
SP2#
SP2#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           1   FULL/BDR        00:00:33    192.168.34.3    FastEthernet1/
0
1.1.1.1           1   FULL/DR         00:00:35    192.168.23.1    FastEthernet0/
0
SP2#
```

```
SP1#ping 3.3.3.3 source 1.1.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 3.3.3.3, timeout is 2 seconds:
Packet sent with a source address of 1.1.1.1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/26/32 ms
SP1#
```

```
SP3#ping 1.1.1.1 source 3.3.3.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 1.1.1.1, timeout is 2 seconds:
Packet sent with a source address of 3.3.3.3
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/27/36 ms
SP3#
```

Étape 3 : EIGRP AS 100 chez le client

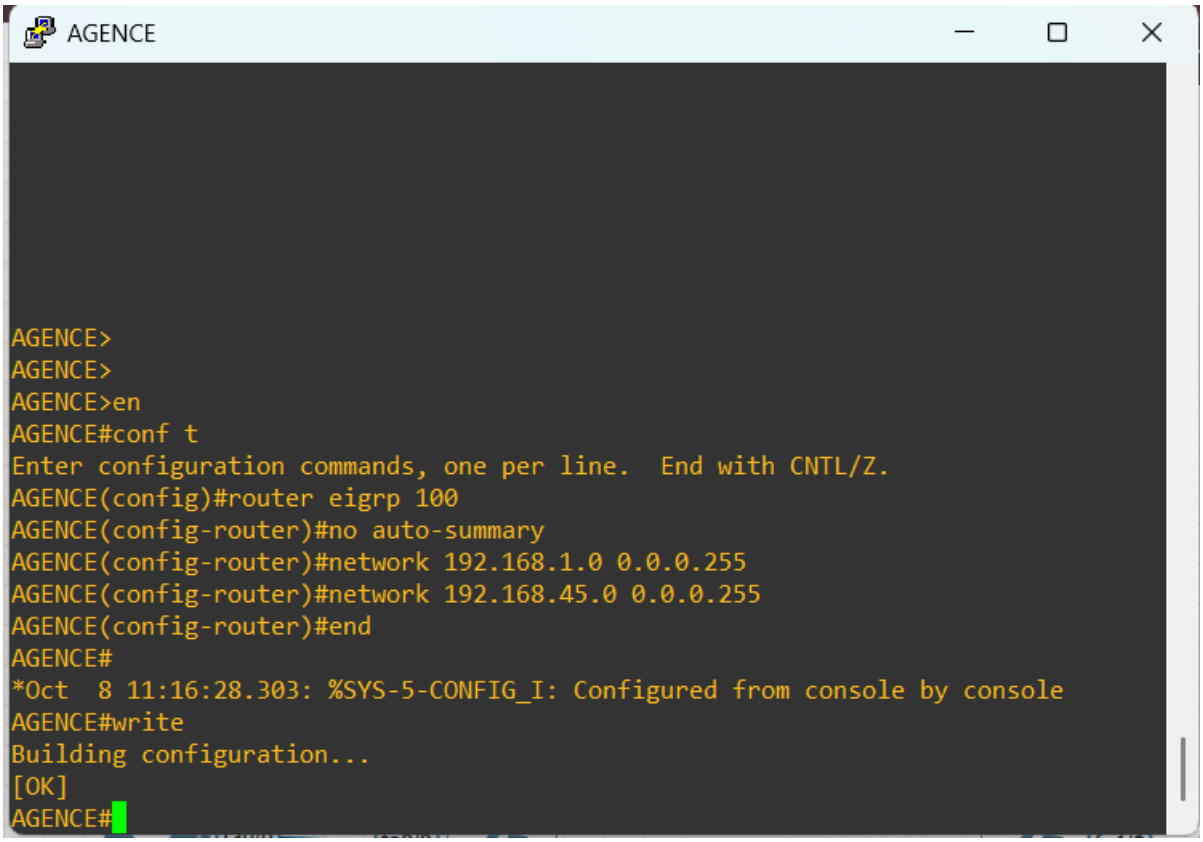
SIEGE

```
SIEGE con0 is now available



Press RETURN to get started.


SIEGE>
SIEGE>en
SIEGE#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SIEGE(config)#router eigrp 100
SIEGE(config-router)#no auto-summary
SIEGE(config-router)#network 10.10.10.0 0.0.0.255
SIEGE(config-router)#network 192.168.12.0 0.0.0.255
SIEGE(config-router)#end
SIEGE#
*Oct  8 11:15:17.751: %SYS-5-CONFIG_I: Configured from console by console
```


AGENCE
AGENCE>
AGENCE>
AGENCE>en
AGENCE#conf t
Enter configuration commands, one per line. End with CNTL/Z.
AGENCE(config)#router eigrp 100
AGENCE(config-router)#no auto-summary
AGENCE(config-router)#network 192.168.1.0 0.0.0.255
AGENCE(config-router)#network 192.168.45.0 0.0.0.255
AGENCE(config-router)#end
AGENCE#
*Oct 8 11:16:28.303: %SYS-5-CONFIG_I: Configured from console by console
AGENCE#write
Building configuration...
[OK]
AGENCE#